KARUNGA JUNIOR SCHOOL
GRADE 9 AGRICULTURE
END OF TERM 1 EXAMINATION

Teacher-prepared assessment resource | Kenya CBC

Learner's Name:	_____	Grade:	9
Date:	_____	Time:	2 hours 30 minutes

INSTRUCTIONS

1. Answer ALL questions.

2. Read each question carefully and give examples where appropriate.

3. Draw labelled diagrams where requested.

4. Observe safety when handling farm tools, materials and animals.

5. The examination carries 100 marks.

SECTION A: SHORT-ANSWER QUESTIONS (40 MARKS)

1. Soil and Soil Fertility (8 marks)

(a) Define soil fertility. (2 marks)

(b) State two benefits of adding organic matter to soil. (2 marks)

(c) Name two types of organic manure. (2 marks)

(d) State two signs that may indicate soil erosion. (2 marks)

2. Crop Production and Seed Selection (8 marks)

(a) State two characteristics of healthy planting materials. (2 marks)

(b) Give two reasons for selecting certified or healthy seeds. (2 marks)

(c) Define seed viability. (2 marks)

(d) State two factors that influence seed germination. (2 marks)

3. Water and Irrigation (8 marks)

(a) State two sources of water used in crop production. (2 marks)

(b) What is irrigation? (2 marks)

(c) Name two methods of irrigation. (2 marks)

(d) State two ways of conserving water on a farm. (2 marks)

4. Crop Management (8 marks)

(a) Explain the meaning of thinning in crop production. (2 marks)

(b) State two reasons for weeding crops. (2 marks)

(c) Name two ways of controlling crop pests. (2 marks)

(d) State two benefits of mulching. (2 marks)

5. Harvesting and Storage (8 marks)

(a) Give two signs that a crop is ready for harvesting. (2 marks)

(b) State two practices that reduce post-harvest losses. (2 marks)

(c) Name two suitable conditions for storing dry grains. (2 marks)

(d) State two reasons for grading agricultural produce. (2 marks)

SECTION B: STRUCTURED QUESTIONS (60 MARKS)

6. Soil Fertility and Conservation (12 marks)

(a) Explain three ways in which organic matter improves soil properties. (6 marks)

(b) Describe three measures a farmer can use to control soil erosion on a sloping farm. (6 marks)

7. Seed Germination and Crop Establishment (12 marks)

(a) State three conditions necessary for successful seed germination. (3 marks)

(b) Describe a simple experiment to investigate the effect of water on seed germination. (5 marks)

(c) Explain two reasons for correct spacing when planting crops. (4 marks)

8. Water Management in Crop Production (12 marks)

(a) Differentiate between surface irrigation and drip irrigation. (4 marks)

(b) Explain three advantages of using drip irrigation in crop production. (6 marks)

(c) State two practices that help prevent water wastage during irrigation. (2 marks)

9. Crop Protection and Field Management (12 marks)

(a) Explain the difference between a crop pest and a crop disease. (4 marks)

(b) Describe four integrated methods a farmer can use to manage crop pests and diseases. (4 marks)

(c) Explain two reasons why farm tools should be cleaned and stored properly after use. (4 marks)

10. Harvesting, Handling and Storage (12 marks)

(a) Describe four steps a farmer should follow when harvesting and handling vegetables for market. (4 marks)

(b) Explain three causes of post-harvest losses. (6 marks)

(c) State two ways of reducing damage by storage pests in dry grains. (2 marks)

MARKING SCHEME
GRADE 9 AGRICULTURE END OF TERM 1

TOTAL: 100 MARKS

1. Soil and Soil Fertility (8 marks)

(a) The ability of soil to supply essential nutrients in suitable amounts and forms to support healthy plant growth. (2)

(b) Any two: improves soil structure; increases water-holding capacity; supports soil organisms; adds nutrients; improves aeration. (2)

(c) Any two: compost manure, farmyard manure, green manure, animal manure. (2)

(d) Any two: rills/gullies; exposed roots; loss of topsoil; sediment deposited downslope; exposed subsoil. (2)

2. Crop Production and Seed Selection (8 marks)

(a) Any two: mature, healthy, undamaged, disease-free, viable and true-to-type. (2)

(b) Any two: better germination; healthy crop stand; uniform growth; reduced disease introduction; reliable variety characteristics. (2)

(c) The ability of a seed to germinate and develop under suitable conditions. (2)

(d) Any two: water/moisture, oxygen, suitable temperature; some seeds also require particular light conditions. (2)

3. Water and Irrigation (8 marks)

(a) Any two: rainfall, rivers, streams, dams, lakes, wells, boreholes, harvested rainwater. (2)

(b) Artificial application of water to soil or crops to support plant growth. (2)

(c) Any two: surface/flood, furrow, sprinkler, drip irrigation. (2)

(d) Any two: mulching; rainwater harvesting; repairing leaks; watering at cooler times; using drip irrigation; covering water storage. (2)

4. Crop Management (8 marks)

(a) Removing excess seedlings so that the remaining plants have adequate space and resources. (2)

(b) Any two: reduces competition for nutrients, water, light and space; removes alternative hosts for pests; improves crop growth and harvesting. (2)

(c) Any two: hand-picking; traps; biological control; crop rotation; resistant varieties; approved pesticides used safely. (2)

(d) Any two: conserves moisture; suppresses weeds; reduces soil erosion; moderates soil temperature; adds organic matter when decomposed. (2)

5. Harvesting and Storage (8 marks)

(a) Any two relevant signs: expected maturity/size/colour; dry mature pods; firm produce; crop-specific maturity indicators. (2)

(b) Any two: harvest at correct maturity; use clean tools/containers; avoid bruising; sort damaged produce; transport carefully; store promptly. (2)

(c) Any two: cool, dry, clean, well-ventilated, pest-free conditions; grain should be adequately dried before storage. (2)

(d) Any two: improve quality consistency; meet market grades; obtain fair prices; separate damaged/poor-quality produce; support suitable packaging. (2)

6. Soil Fertility and Conservation (12 marks)

(a) Any three explained: adds nutrients; improves aggregation/structure; increases water retention; supports beneficial organisms; improves aeration and root penetration. Award 2 each. (6)

(b) Any three described: contour farming/ploughing; terracing; planting grass strips; cover crops; mulching; agroforestry; controlled drainage; maintaining vegetation. Award 2 each. (6)

7. Seed Germination and Crop Establishment (12 marks)

(a) Water, oxygen and suitable temperature. (3)

(b) Place equal numbers of similar seeds on moist cotton/paper in one container and dry cotton/paper in another; keep other conditions similar; observe daily; record number germinating; compare results. Award up to 5 for fair test, setup, control, observation and conclusion.

(c) Any two explained: reduces competition; allows air/light penetration; supports healthy root and shoot development; eases weeding and harvesting; reduces disease spread. (4)

8. Water Management in Crop Production (12 marks)

(a) Surface irrigation allows water to flow over the soil surface (e.g. furrows); drip irrigation delivers small amounts directly near plant roots through emitters/tubes. (4)

(b) Any three explained: reduces evaporation/runoff; directs water to root zone; saves water; reduces weed growth between rows; can support uniform application; keeps foliage drier and may reduce some diseases. (6)

(c) Any two: repair leaks; irrigate when evaporation is low; use soil-moisture checks; avoid overwatering; maintain pipes and emitters. (2)

9. Crop Protection and Field Management (12 marks)

(a) A pest is an organism (e.g. insect, rodent or bird) that damages crops; a disease is a harmful condition caused by pathogens or other disease agents, often producing symptoms such as spots, wilting or rotting. (4)

(b) Any four: field sanitation; crop rotation; resistant varieties; biological control; traps; hand removal; correct spacing; removal of infected plants; safe use of approved pesticides. (4)

(c) Any two explained: prevents rust/corrosion; reduces spread of pests and diseases; keeps tools effective; prevents accidents; extends tool life. (4)

10. Harvesting, Handling and Storage (12 marks)

(a) Any four: harvest at correct maturity; use clean/sharp tools; avoid bruising; keep produce shaded; sort and grade; use clean ventilated containers; transport promptly and carefully. (4)

(b) Any three explained: poor handling/bruising; pests and rodents; fungal/bacterial decay; excess moisture; unsuitable temperature; delayed transport/storage; harvesting at wrong maturity. (6)

(c) Any two: dry grain to safe moisture level; clean and disinfect storage area appropriately; use airtight/hermetic or suitable pest-proof storage; inspect regularly; keep stores clean and dry. (2)

GENERAL MARKING GUIDANCE

Accept other scientifically and agriculturally correct responses relevant to the question.

Award marks for clear explanations, correct sequences and practical understanding.

For experiment questions, reward a fair test, control of variables, accurate observation and a valid conclusion.

Teachers should confirm that the assessed sub-strands have been taught and adapt local examples to the crops and farming practices in the learners' area.

This is an original teacher-prepared assessment resource and is not an official KICD examination.

END OF EXAMINATION